

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Sampling techniques & measuring abiotic factors - Set 1

Mark scheme

AQA spec 3.7.2.1-3.7.2.2; Research methods; textbook pp. 394-419

Marking guidance

- Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
- The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
- **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	1 mark each 1 mark for each of the 2 points below. 1 Quadrat size depends on the size / spacing of the organisms: 0.5 m x 0.5 m is typical for ground flora 1 10 cm x 10 cm is for lichens (too small to contain a representative sample of plants) – 100 m x 100 m is for trees (impractical, too much work) Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • quadrat → a square frame of known area used to sample a fixed area	2

Question	Marking guidance	Marks
01.2	Calculation A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. 1 $N = 240$ – $N(N-1) = 240 \times 239 = 57\,360$ 1 $\Sigma n(n-1) = 48 \times 47 + 120 \times 119 + 36 \times 35 + 24 \times 23 + 12 \times 11 = 2256 + 14280 + 1260 + 552 + 132 = 18\,480$ 1 $D = 57\,360 / 18\,480 = 3.10$ (accept 3.1) Also accept • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • Simpson → a diversity index that uses both the number of species and their abundance • Biodiversity → the variety of life: habitat, species and genetic diversity Do not credit • A bare wrong answer with no working. • 'more species always means a higher index' - the evenness of the abundances matters as well	3
01.3	1 mark each 1 mark for each of the 2 points below. 1 Lower D means lower biodiversity: the ungrazed meadow is dominated by one or a few species (eg tall grass) with the others rare (less even) 1 Grazing (plagioclimax management) prevents dominance by vigorous species so abundance is more evenly spread Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • biodiversity → the variety of life: habitat, species and genetic diversity • plagioclimax → a community held back from its climax by human activity	2
01.4	1 mark each 1 mark for each of the 2 points below. 1 A frame quadrat flattens tall plants, making percentage cover estimates inaccurate and hiding shorter plants 1 Point quadrat pins are lowered through the vegetation – each touch counts (10 pins = 10% each) without flattening Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • quadrat → a square frame of known area used to sample a fixed area	2
01.5	1 mark for the point below. 1 Subjective judgement varies between recorders – results are not quantitative so statistical analysis is not possible (but quick for many areas) Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 394-395, 401-402; AQA spec 3.7.2.1.	

Question	Marking guidance	Marks
02.1	1 mark each 1 mark for each of the 3 points below. 1 Net held with its straight edge on the river bed facing the current (open to the flow) 1 The bed in front is disturbed by kicking / scraping a boot for a standard number of kicks and time (eg 30 s) 1 Same substrate type, water depth and flow rate at both sites – same net mesh – repeat samples – invertebrates identified with a key and counted Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • kick sampling → disturbing the stream bed and catching the organisms carried downstream	3
02.2	Any 3 of 5 1 mark each for any 3 of the 5 points below, in any order, maximum 3. 1 Difficult to standardise the vigour / area kicked 1 Buried organisms or those fixed to stones are not dislodged 1 Some organisms swim and avoid the net 1 Flow rate varies and affects capture 1 Fragile organisms destroyed Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • kick sampling → disturbing the stream bed and catching the organisms carried downstream Do not credit • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
02.3	1 mark each 1 mark for each of the 2 points below. 1 A fixed area of bed is enclosed by a frame so the sampled area is standardised – side nets reduce escapes 1 But only works in fast enough flow and is hard to set on rough substrates / deep water Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • kick sampling → disturbing the stream bed and catching the organisms carried downstream	2

Question	Marking guidance	Marks
02.4	1 mark each 1 mark for each of the 2 points below. 1 Pollutant / oxygen levels fluctuate (eg with flow, time of day) so one reading may miss a pollution event 1 Sensitive taxa (stonefly, mayfly larvae) disappear and tolerant ones (midge larvae, hoglouse) remain, integrating conditions over weeks / months (biotic index) Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. biotic → living factors / effects of other organisms	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 395-396, 415-416; AQA spec 3.7.2.1.	
03.1	2 marks per point 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. 2 Ground surface: pitfall trap (container sunk level with the surface, cover, checked after 24 h) - only mobile surface animals - trapped predators eat others - preservative attracts / repels - escapes 2 Leaf litter: Tullgren funnel (sample on mesh under a light; heat / light drive invertebrates down into a container for a standard time) - only species repelled by heat / light - some die or stop moving - large ones cannot pass the mesh 2 Branches: beating tray (branch beaten a set number of times over a sheet) - hard to standardise vigour - some species not dislodged / fly away - high branches unreachable Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. pitfall trap → a sunken container that catches ground invertebrates Tullgren → a funnel that drives soil invertebrates down into a collecting jar	6
03.2	1 mark each 1 mark for each of the 2 points below. 1 Moths only fly as adults for part of the year and activity depends on weather / season, so a species may be present but not caught 1 Brightness / wavelength of the lamp affects attractiveness so catches vary between traps Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Question	Marking guidance	Marks
03.3	1 mark each 1 mark for each of the 2 points below. 1 The area of ground covered by the suction pipe is known so invertebrates per m² can be calculated 1 A pooter is only a handling / collection device for individuals already found, not a sampling method Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. - The converse of any point, ie the same comparison written the other way round.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 397-399; AQA spec 3.7.2.1.	
04.1	Calculation A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. 1 $(60 \times 45) / 9$ 1 = 300 Also accept - The correct answer however it is set out, including a different but valid method. - An answer carried forward from an earlier incorrect step (error carried forward). - An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. - The value alone - the unit (beetles) is already printed on the answer line. Do not credit - A bare wrong answer with no working.	2
04.2	Any 4 of 5 1 mark each for any 4 of the 5 points below, in any order, maximum 4. 1 Marked individuals mix freely / randomly with the rest of the population 1 No births, deaths, immigration or emigration between samples 1 Marking does not affect survival or behaviour (eg make them more visible to predators) 1 Marks do not wear off / fall off 1 All individuals have the same chance of being caught (traps do not attract or repel marked animals) Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. Lincoln Index → estimating population size from the proportion of marked animals recaptured Do not credit - Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. 'pick places at random by eye' - positions have to be generated with random numbers to remove bias	4

Question	Marking guidance	Marks
04.3	1 mark each 1 mark for each of the 2 points below. 1 Fewer marked beetles would survive to be recaptured 1 So the proportion marked in the second sample would be too low and the population would be overestimated Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	2
04.4	1 mark each 1 mark for each of the 2 points below. 1 Enough time for marked beetles to mix with the population 1 Not so long that births / deaths / migration or loss of marks change the population Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 401-402; AQA spec 3.7.2.1.	
05.1	1 mark each 1 mark for each of the 3 points below. 1 Readings drift so they may not be accurate - calibration adjusts the meter to match a known standard so results are accurate and comparable between meters / occasions 1 pH meter placed in buffer solutions of known pH (eg 4, 7, 10) and adjusted to read correctly 1 If a meter cannot be adjusted, record the percentage error and correct the results Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. accurate → close to the true value	3
05.2	1 mark each 1 mark for each of the 3 points below. 1 Same height above the ground 1 No obstacles to air flow - impeller type facing into the wind 1 Axis of rotation horizontal (impeller) or vertical (cup) - same time period / averaging Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes.	3
05.3	1 mark each 1 mark for each of the 2 points below. 1 Two thermometers, one with a wet bulb, are spun until the wet bulb stops cooling 1 Evaporation cools the wet bulb more when humidity is low - the temperature difference is compared with a table to find humidity Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Question	Marking guidance	Marks
05.4	Any 2 of 3 1 mark each for any 2 of the 3 points below, in any order, maximum 2. 1 Test strips with reagent bands compared with a colour chart (semi-quantitative) 1 Electronic colourimeter: reagent added and colour intensity measured at a specific wavelength 1 Ion-selective electrode measuring electrode potential (other ions can interfere) Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. Do not credit - Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 407-410, 413; AQA spec 3.7.2.1.	
06.1	1 mark each 1 mark for each of the 3 points below. 1 Sieves assembled with the coarsest mesh on top – a dried, crushed soil sample is placed in the top sieve 1 Stack shaken (about 2 minutes) until no more particles pass – each fraction weighed 1 Percentages of sand (0.02-2 mm), silt (0.002-0.02 mm) and clay (base pan, below 0.002 mm) calculated – material over 2 mm (stones, twigs) is not soil – plotted on a soil triangle Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	3
06.2	1 mark each 1 mark for each of the 2 points below. 1 Auger collects a core of known cross-sectional area and depth 1 So samples from different sites are directly comparable (also needed for bulk density = dry mass / volume) Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	2
06.3	1 mark each 1 mark for each of the 2 points below. 1 Minerals such as calcium carbonate break down releasing carbon dioxide 1 So mass is lost that is not organic matter Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Question	Marking guidance	Marks
06.4	1 mark each 1 mark for each of the 3 points below. 1 A beam of light passes through the sample in a dark chamber – a light sensor at 90° measures light scattered by suspended particles - more turbid, more scatter 1 Standardise bulb brightness / colour, distance and container 1 A calibration curve is produced using samples with known amounts of suspended matter so readings convert to concentrations – Secchi discs / turbidity bottles give only semi-quantitative results Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 410-413, 417; AQA spec 3.7.2.1-3.7.2.2.	
07.1	1 mark each 1 mark for each of the 2 points below. 1 Small seeds, genetically uniform, high and rapid germination rate so results are quick and reliable 1 Wild species have different ranges of tolerance so cress results cannot be applied directly Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. reliable → repeatable / gives consistent results when repeated tolerance → the range of conditions in which a species can survive Do not credit 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	2
07.2	1 mark each 1 mark for each of the 2 points below. 1 Preliminary study with widely spaced pH values to find where germination occurs 1 Final range from below the lowest pH giving germination to above the highest, with closer intervals Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	2
07.3	1 mark each 1 mark for each of the 4 points below. 1 Temperature (constant, natural germination temperature) 1 Light, water volume, number of seeds per dish, seed source 1 Growth medium (filter paper; soil would alter pH) – nutrients not needed for germination 1 Continue until it is clear that ungerminated seeds will not germinate, so that a delay in germination is not mistaken for failure Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	4

Question	Marking guidance	Marks
07.4	1 mark each 1 mark for each of the 2 points below. 1 Surface water affects fresh mass and drying damages the plants – dry mass requires killing the plants so the same plants cannot be re-measured 1 Leaf area is a reasonable estimate of biomass that allows repeated measurement of the same plants over time Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 416-418; AQA spec 3.7.2.2.	